

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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Day 5 — Partial, Access, Strings, Records, Indexers + Mini-Projects

Task 1.17 Partial class Employee across two files with a partial method; an access-modifier matrix class; Address (Street, City, Pincode) record compared with == and copied with; a Playlist indexer with bounds checking + a string indexer. ✓ Done when: the record prints True for value equality and with produces a changed copy.

Procedure:

I learnt what is partial method, how record is used for value comparison between two objects, and how to modify a single value of object using keyword. Using an indexer to access objects like arrays.

```

Program.cs
HelloCohort
Program
Main()

1 // Record class
2 public record Address(string Street, string City, int Pincode);
3
4 // INDEXER CLASS
5 class Playlist
6 {
7     private string[] songs = new string[10];
8
9     // INDEXER
10    public string this[int index]
11    {
12        get
13        {
14            if (index < 0 || index >= songs.Length)
15                return "Invalid index";
16            return songs[index];
17        }
18        set
19        {
20            if (index < 0 || index >= songs.Length)
21                songs[index] = null;
22            songs[index] = value;
23        }
24    }
25 }
26
27 // PARTIAL CLASS (single file demo)
28 public partial class Employee
29 {
30     public string Name { get; set; }
31
32     // METHOD
33     partial void ShowMessage();
34
35     // METHOD
36     public void Display()
37     {
38         Console.WriteLine(Name);
39         ShowMessage();
40     }
41 }
42
43 public partial class Employee
44 {
45     // METHOD
46     partial void ShowMessage()
47     {
48         Console.WriteLine("Hello from partial method");
49     }
50 }
51
52 // PROGRAM
53 class Program
54 {
55     // METHOD
56     static void Main()
57     {
58         // PARTIAL CLASS
59         Employee e = new Employee { Name = "Ravi" };
60         e.Display();
61
62         // RECORD
63         var a1 = new Address("Main", "Chennai", 600001);
64         var a2 = new Address("Main", "Chennai", 600001);
65         Console.WriteLine(a1 == a2); // True
66
67         var a3 = a1 with { City = "Mumbai" };
68         Console.WriteLine(a3);
69
70         // INDEXER
71         Playlist p = new Playlist();
72         p[0] = "Song1";
73         p[1] = "Song2";
74         Console.WriteLine(p[0]);
75         Console.WriteLine(p[5]); // Invalid index
76     }
77 }
78
79 60% 0 1 ↑ ↓ ↶ ↷ Lrn 76, Ch: 1 SPC CRLF UTF-8 with BOM

```

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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```

Microsoft Visual Studio Debug Console
Ravi
Hello from partial method
True
Address { Street = Main, City = Mumbai, Pincode = 600001 }
Song1
Invalid index

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 28752) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Mini Q1 — Product Catalog: `Product {Name, Price, Category(enum) }`, 5 objects printed grouped by category. ✓ Done when: output is grouped correctly by category.

Procedure:

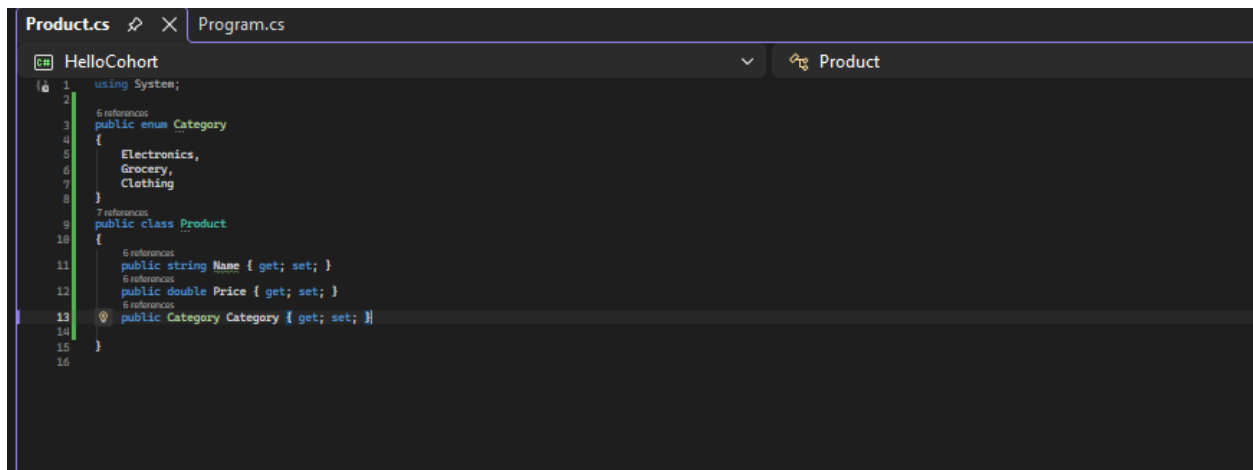
Displaying the categorized product catalog using Linq groupBy().

```

Product.cs | Program.cs
HelloCohort
7 // Class Program
8 {
9     // References
10    static void Main()
11    {
12        // Create products
13        List<Product> products = new List<Product>
14        {
15            new Product { Name = "Laptop", Price = 50000, Category = Category.Electronics },
16            new Product { Name = "Phone", Price = 20000, Category = Category.Electronics },
17            new Product { Name = "Shirt", Price = 1000, Category = Category.Clothing },
18            new Product { Name = "Rice", Price = 50, Category = Category.Grocery },
19            new Product { Name = "Jeans", Price = 2000, Category = Category.Clothing }
20        };
21
22        // Group by category
23        var groupedProducts = products.GroupBy(p => p.Category);
24
25        // Print grouped output
26        foreach (var group in groupedProducts)
27        {
28            Console.WriteLine($"Category: {group.Key}");
29
30            foreach (var product in group)
31            {
32                Console.WriteLine($"  {product.Name} - ₹{product.Price}");
33            }
34
35            Console.WriteLine();
36        }
37    }
38 }
39
40
41
42

```

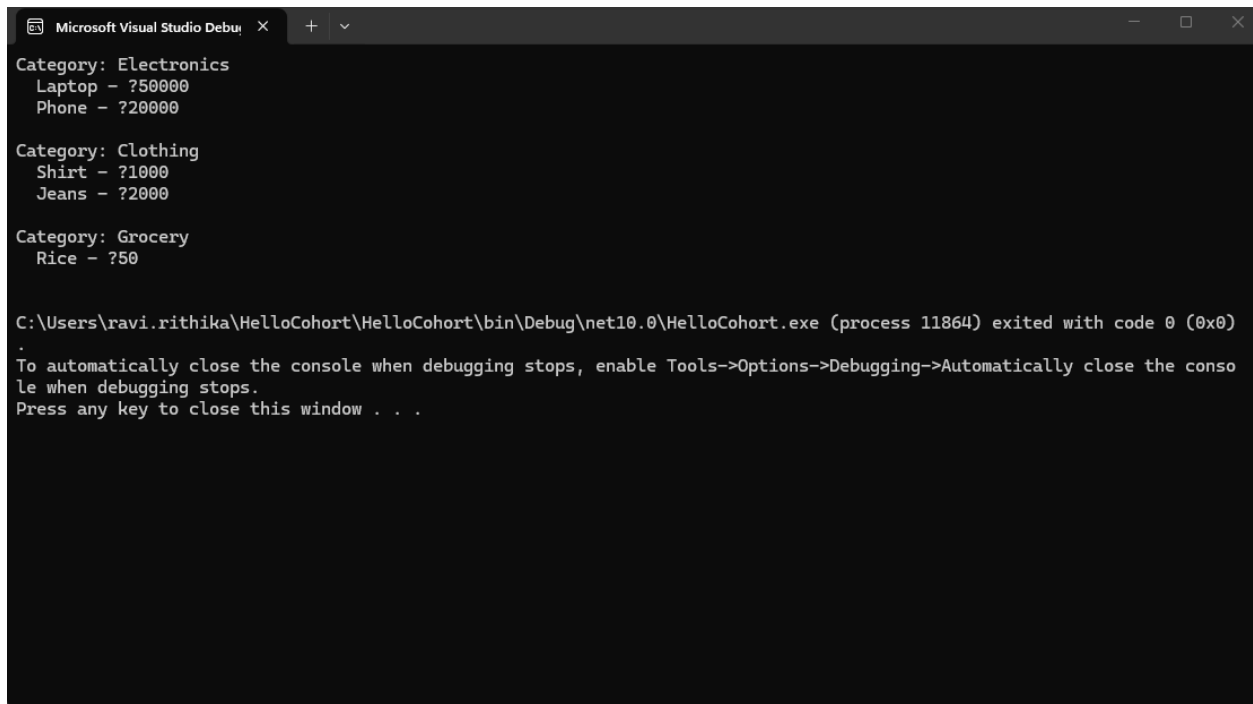
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```

1 using System;
2
3 6 references
4 public enum Category
5 {
6     Electronics,
7     Grocery,
8     Clothing
9 }
10 7 references
11 public class Product
12 {
13     6 references
14     public string Name { get; set; }
15     6 references
16     public double Price { get; set; }
17     6 references
18     public Category Category { get; set; }
19 }
20

```



```

Category: Electronics
Laptop - ₹50000
Phone - ₹20000

Category: Clothing
Shirt - ₹1000
Jeans - ₹2000

Category: Grocery
Rice - ₹50

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 11864) exited with code 0 (0x0)
.
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Press any key to close this window . . .

```

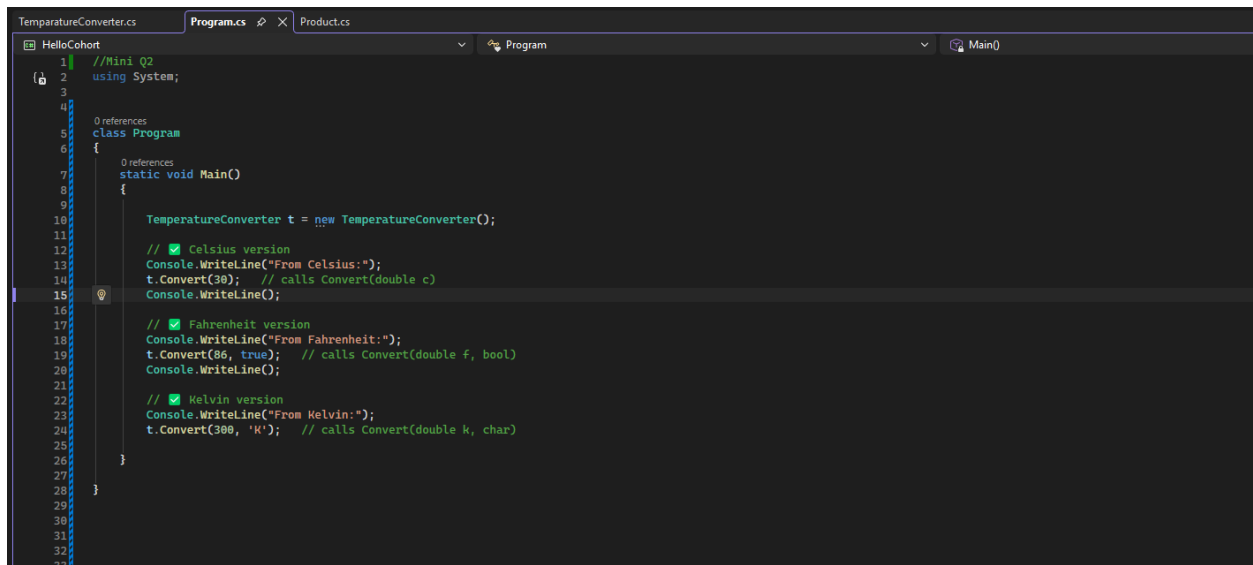
Mini Q2 — Temperature Converter: value + unit → the other two units, using method overloading. ✓
 Done when: all three units convert correctly.

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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Procedure:

Conversion formulae $F = (C \times 9/5) + 32$ and $K = C + 273.15$

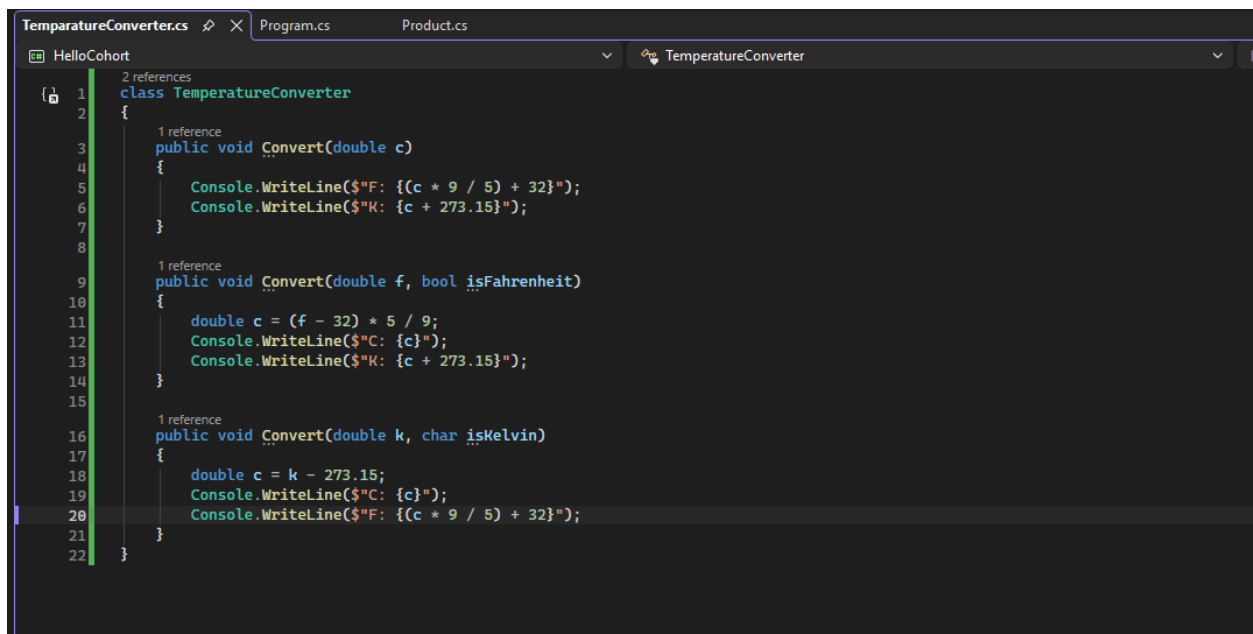
Temperature Converter using method overloading.



```

1 //Mini Q2
2 using System;
3
4
5 0 references
6 class Program
7 {
8     0 references
9     static void Main()
10    {
11        TemperatureConverter t = new TemperatureConverter();
12
13        // Celsius version
14        Console.WriteLine("From Celsius:");
15        t.Convert(30); // calls Convert(double c)
16        Console.WriteLine();
17
18        // Fahrenheit version
19        Console.WriteLine("From Fahrenheit:");
20        t.Convert(86, true); // calls Convert(double f, bool)
21        Console.WriteLine();
22
23        // Kelvin version
24        Console.WriteLine("From Kelvin:");
25        t.Convert(300, 'K'); // calls Convert(double k, char)
26    }
27 }
28
29
30
31
32
33

```

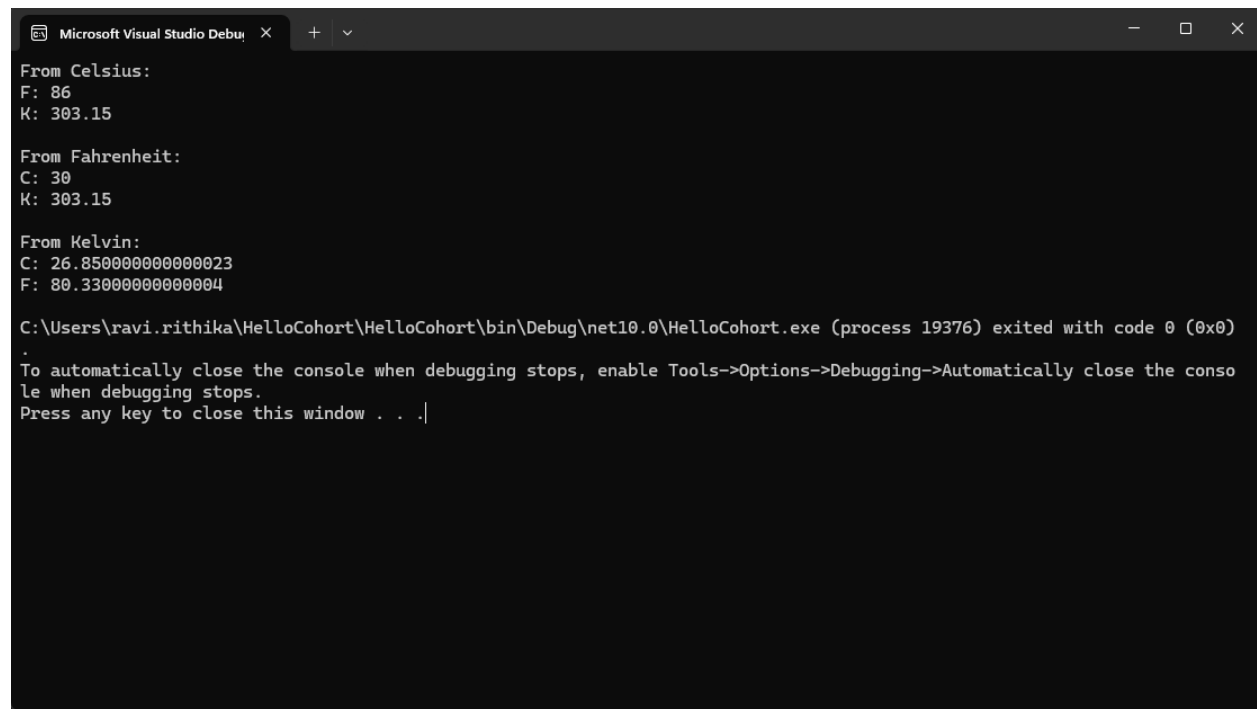


```

1 2 references
2 class TemperatureConverter
3 {
4     1 reference
5     public void Convert(double c)
6     {
7         Console.WriteLine($"F: {(c * 9 / 5) + 32}");
8         Console.WriteLine($"K: {c + 273.15}");
9     }
10
11     1 reference
12     public void Convert(double f, bool isFahrenheit)
13     {
14         double c = (f - 32) * 5 / 9;
15         Console.WriteLine($"C: {c}");
16         Console.WriteLine($"K: {c + 273.15}");
17     }
18
19     1 reference
20     public void Convert(double k, char isKelvin)
21     {
22         double c = k - 273.15;
23         Console.WriteLine($"C: {c}");
24         Console.WriteLine($"F: {(c * 9 / 5) + 32}");
25     }
26 }
27
28
29
30
31
32

```

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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```

Microsoft Visual Studio Debug Console
+ -

From Celsius:
F: 86
K: 303.15

From Fahrenheit:
C: 30
K: 303.15

From Kelvin:
C: 26.8500000000000023
F: 80.330000000000004

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 19376) exited with code 0 (0x0)
.
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Press any key to close this window . . .|

```

Mini Q3 — Contact Card: ContactCard struct array of 5; case-insensitive search by name. ✓ Done when: a lowercase query finds a differently cased name.

Procedure:

I have created a ContactCard struct and implemented a name search which is case-insensitive. By converting the passed value to lowercase using in-built .ToLower() method.

Rithika Ravi	P575	B01	"Week 1 — C# Language & Object Foundations".
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```

1  using System;
2  8 references
3  struct ContactCard
4  {
5      public string Name;
6      public string Phone;
7  }
8  0 references
9  class Program
10 {
11     0 references
12     static void Main()
13     {
14         ContactCard[] contacts = new ContactCard[5];
15
16         contacts[0] = new ContactCard { Name = "Ravi", Phone = "111" };
17         contacts[1] = new ContactCard { Name = "Anita", Phone = "222" };
18         contacts[2] = new ContactCard { Name = "RAHUL", Phone = "333" };
19         contacts[3] = new ContactCard { Name = "Meena", Phone = "444" };
20         contacts[4] = new ContactCard { Name = "Kiran", Phone = "555" };
21
22         // Call search
23         SearchContact(contacts, "rahul");
24     }
25     1 reference
26     static void SearchContact(ContactCard[] contacts, string searchName)
27     {
28         searchName = searchName.ToLower();
29
30         foreach (var contact in contacts)
31         {
32             if (contact.Name.ToLower() == searchName)
33             {
34                 Console.WriteLine($"Found: {contact.Name} - {contact.Phone}");
35                 return;
36             }
37         }
38
39         Console.WriteLine("Contact not found");
40     }
41 }
42

```

```

Microsoft Visual Studio Debug Console
Found: RAHUL - 333

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 17512) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```